

A Unified Machine Learning Framework

Customer Segmentation and Product Preference Prediction
in Retail Analytics

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The Challenge

Traditional Approach

Static attributes and gut instinct are **no longer sufficient** for customer understanding

Market Need

Move from **intuition-based** to **data-driven predictions** for personalized experiences

Research Gap

Few studies link behavioral customer segmentation with predictive modeling of product preference



Literature Review

Research Foundations

Existing Research vs. Our Approach

Current Research

Methods Used

K-Means, DBSCAN, SVM, Random Forest, XGBoost for general segmentation (Ramkumar et al. 2025; Lathifah & Azzahra, 2025)

Focus Areas

Churn prediction, loyalty tiers, broad outcomes - not specific product preferences (Kabir et al., 2025)

Limitations

Heavy reliance on RFM features, ignoring multi-category spending behaviors (Bhatia & Gupta 2025; Nasirzonouzi, 2025)

Our Innovation

Unified Framework

Integrate unsupervised clustering with supervised learning for comprehensive insights

Real-World Data

Full behavioral, transactional, and marketing campaign features from retail dataset

Product Prediction

Multi-class classification framework for next likely product purchase



Two-Stage Approach

1

Unsupervised Clustering

K-Means, DBSCAN, Agglomerative clustering for customer segmentation

2

Supervised Learning

Logistic Regression, Random Forest, XGBoost, k-NN and Decision Tree for product prediction

3

Integration

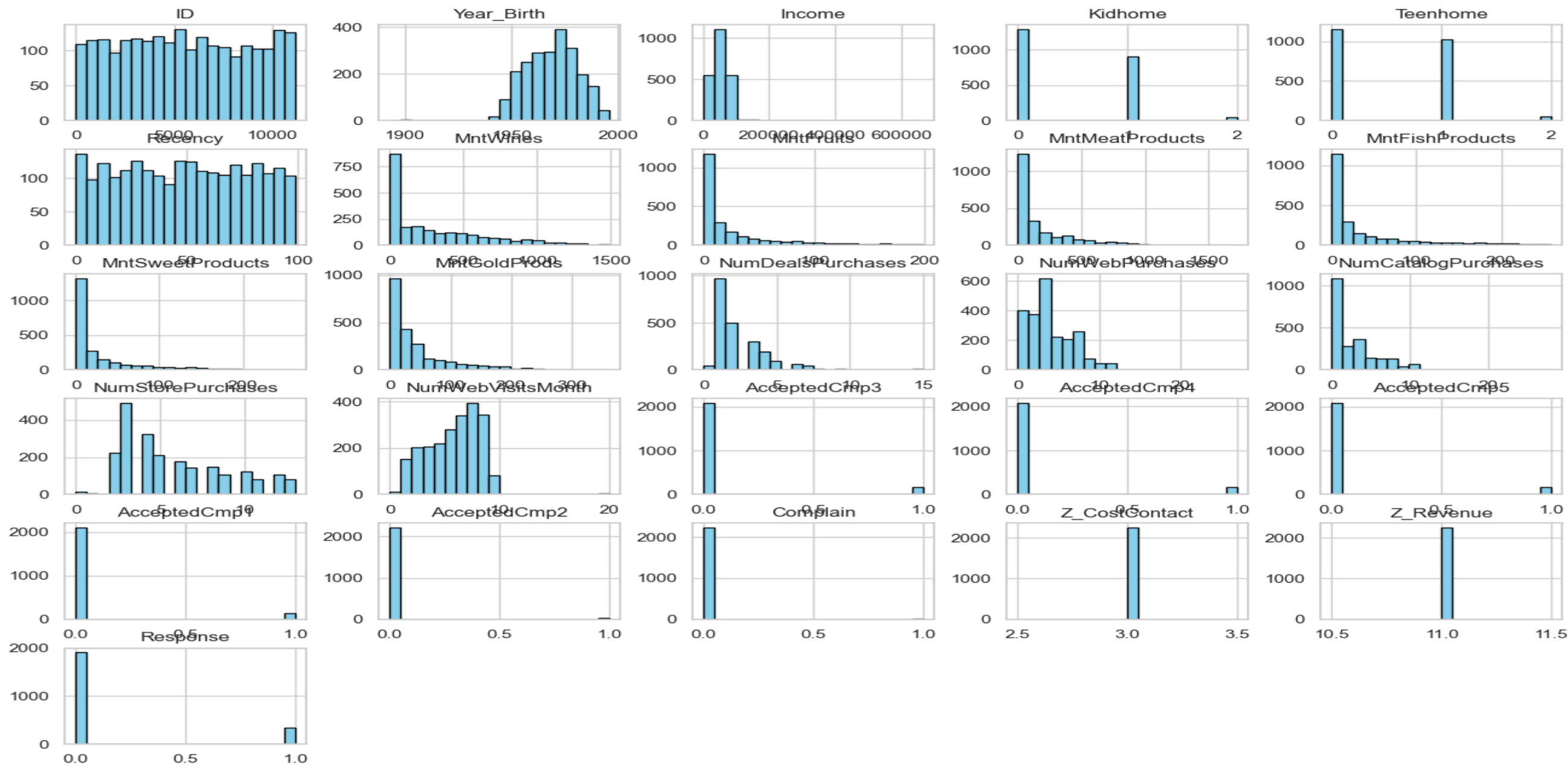
Combine segments with predictive models for actionable insights

Unified ML Framework



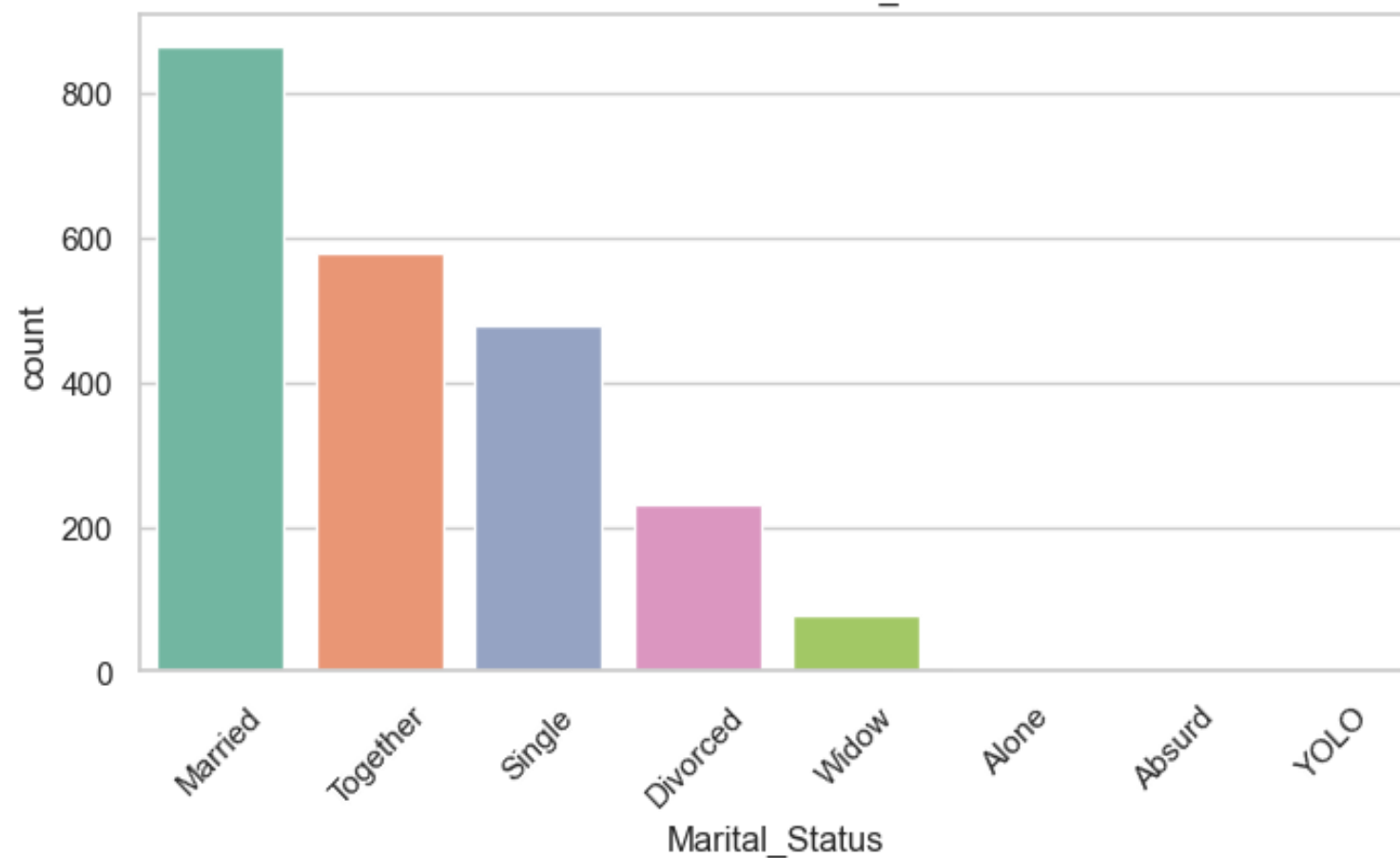
Exploratory Data Analysis

Distribution of Numerical Variables

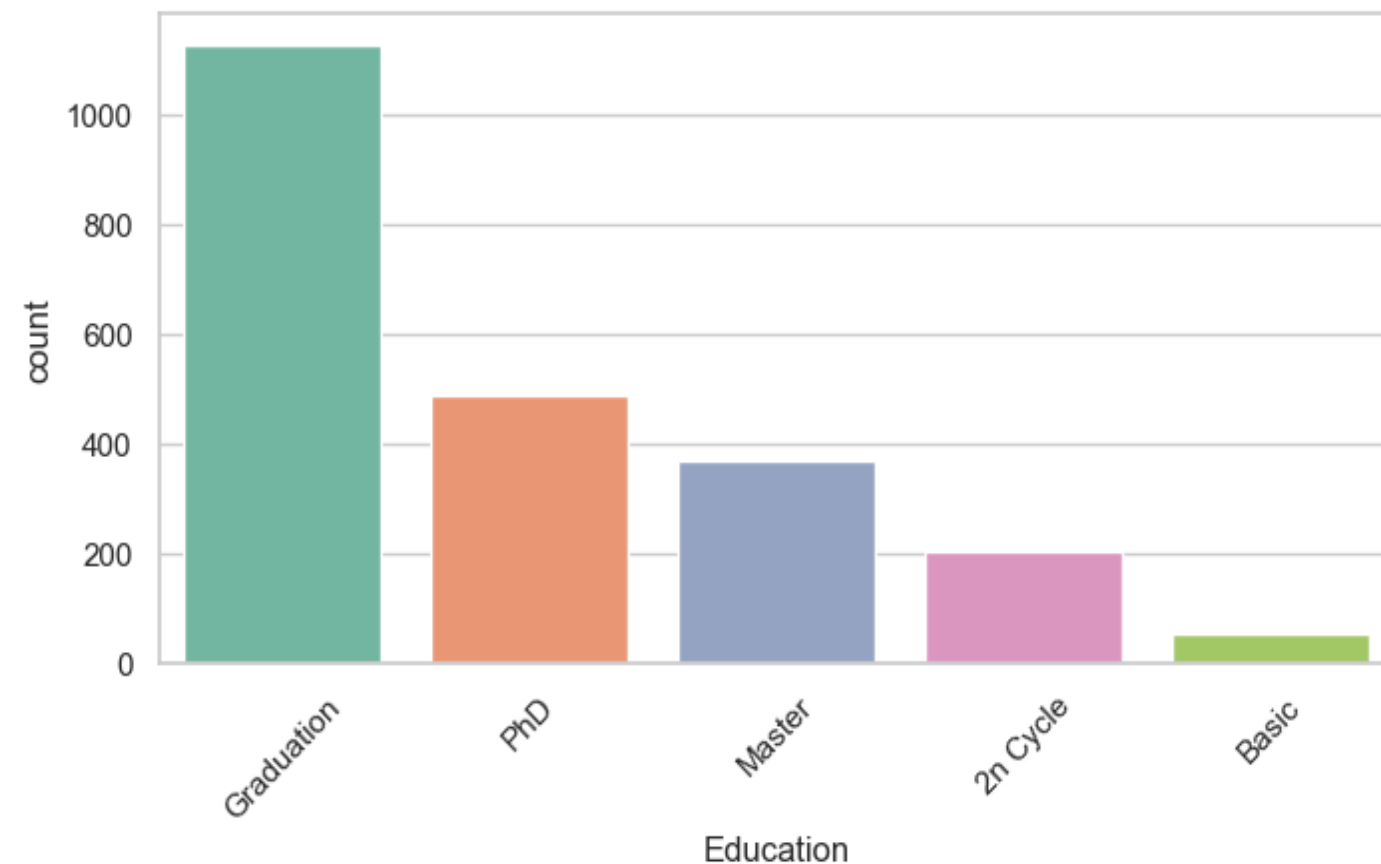


Exploratory Data Analysis

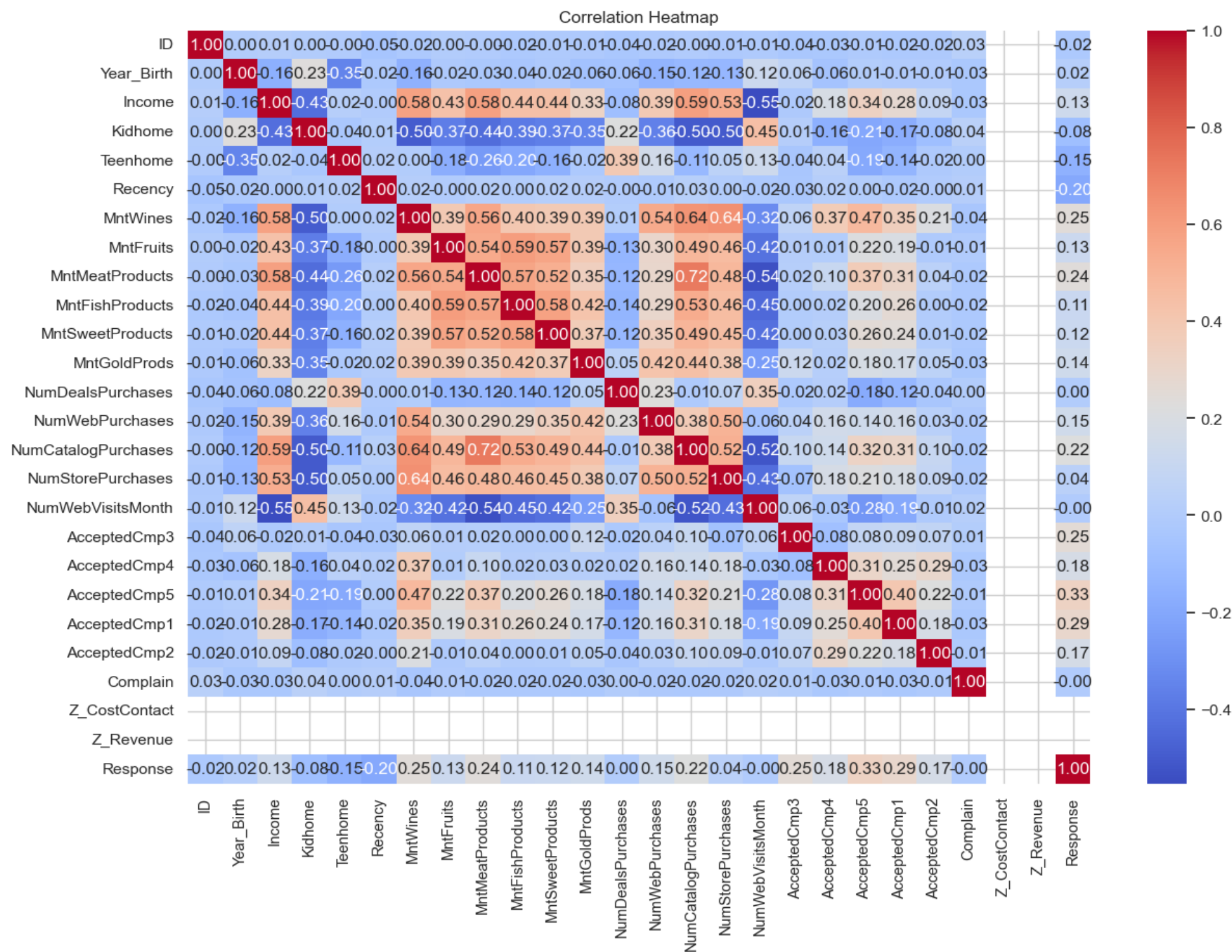
Distribution of Marital_Status



Distribution of Education



Correlation Heatmap



Data preprocessing

Data cleaning



Step 2

Data - Transformative

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Data Preprocessing Pipeline

Feature Selection

Dropped irrelevant columns including IDs, dates, and spending targets to prevent data leakage

Missing Value Treatment

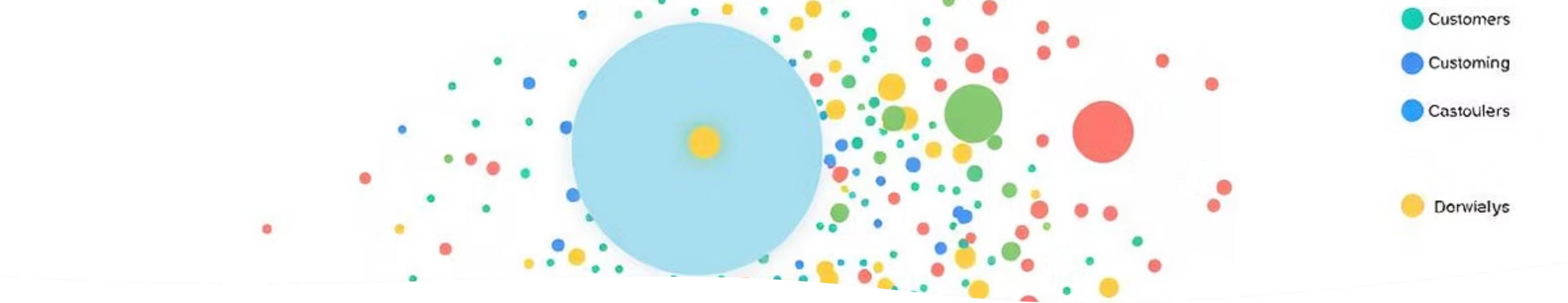
Applied imputation strategies for key columns like Income using appropriate statistical methods

Categorical Encoding

One-hot encoding applied to Education and Marital_Status for ML compatibility

Feature Scaling

StandardScaler applied to all continuous variables for uniform feature scaling

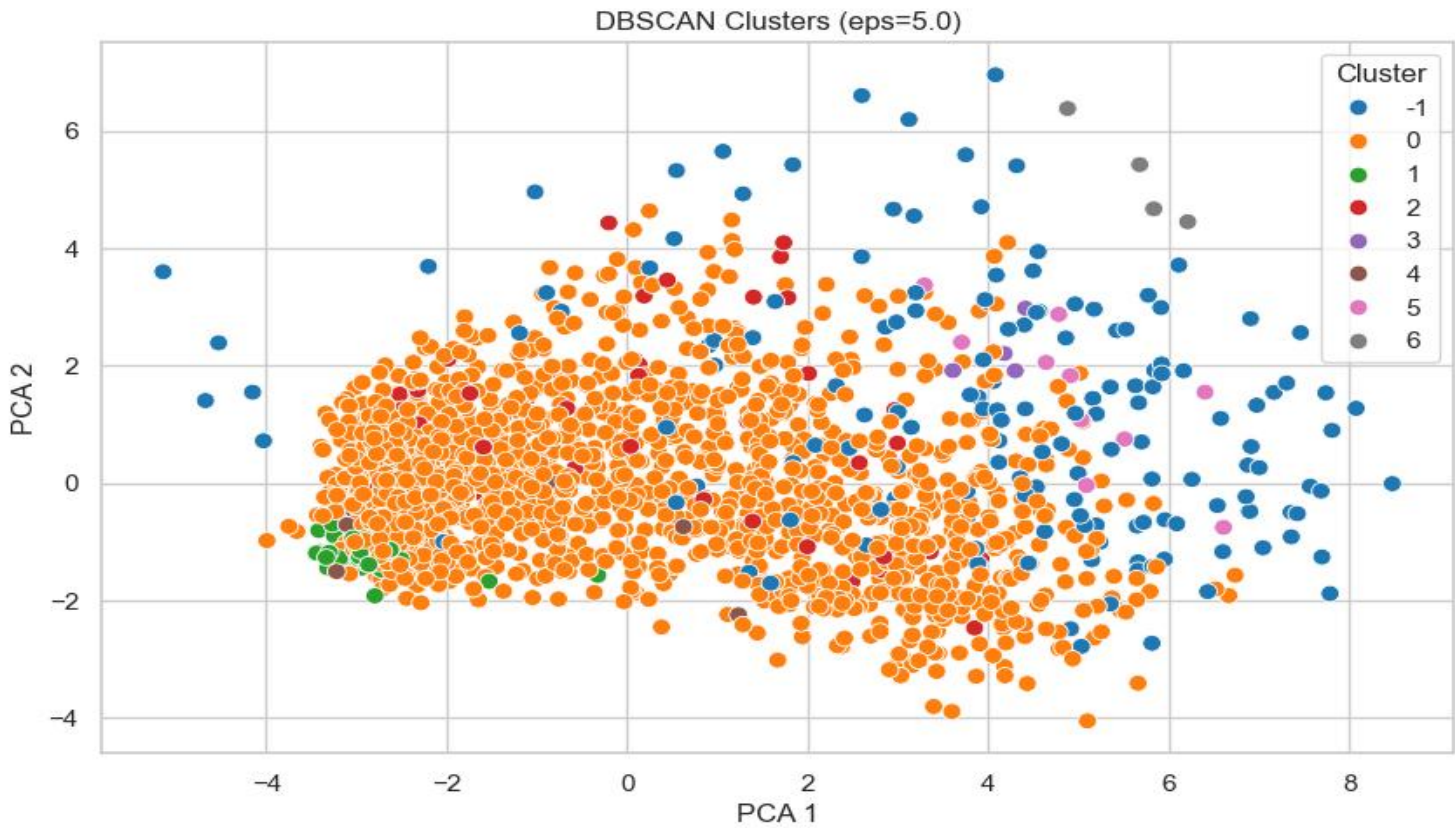
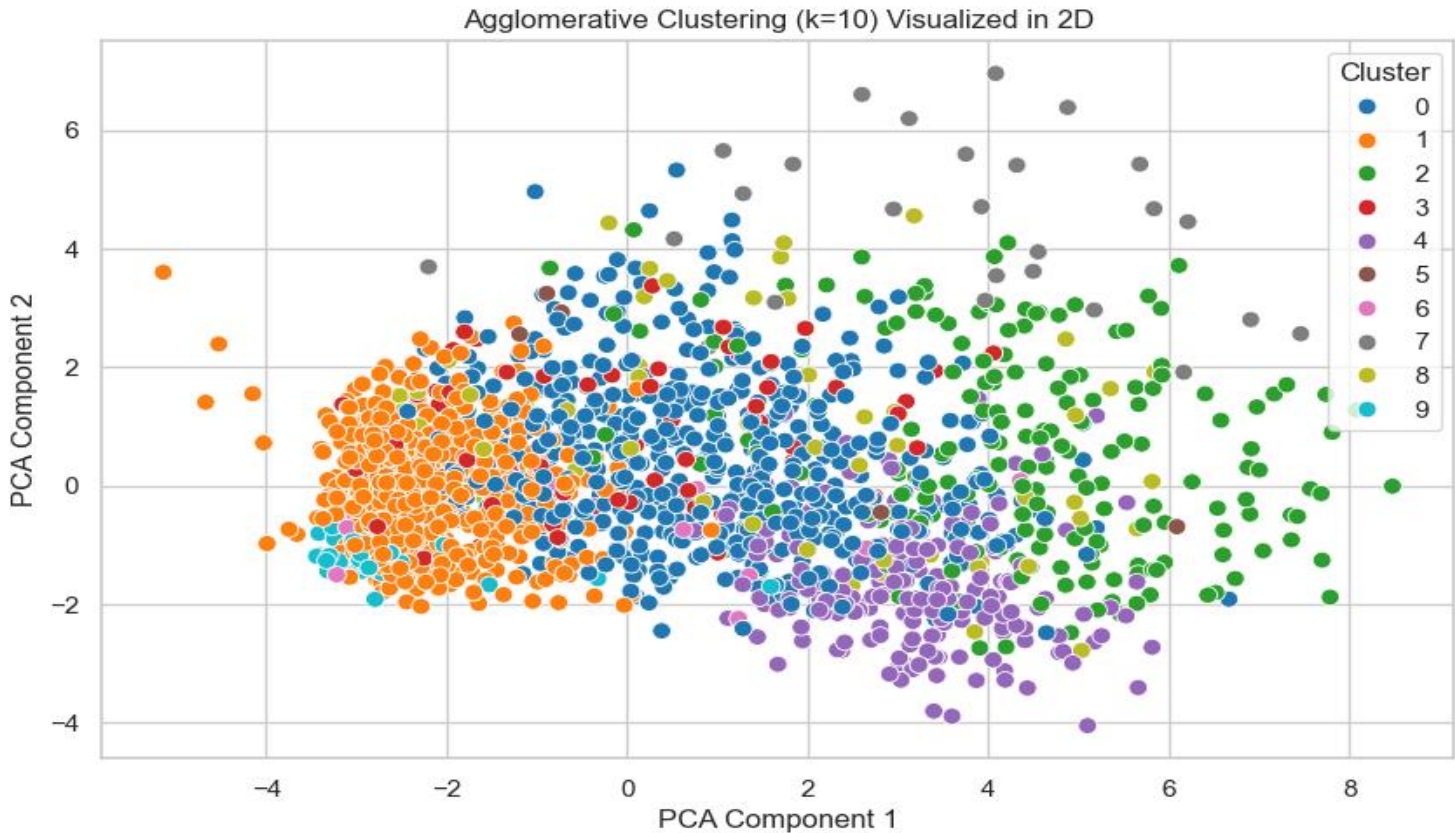
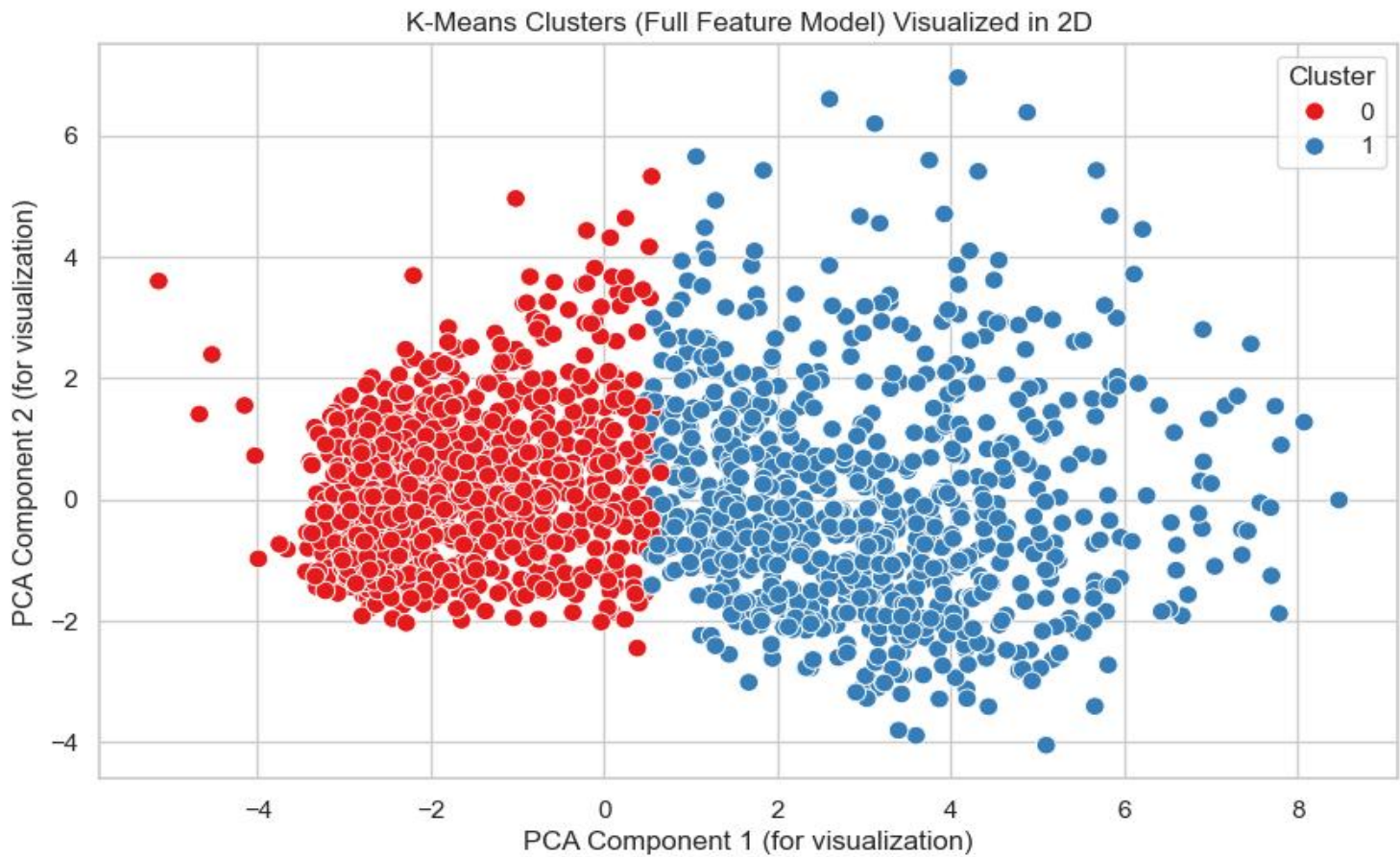


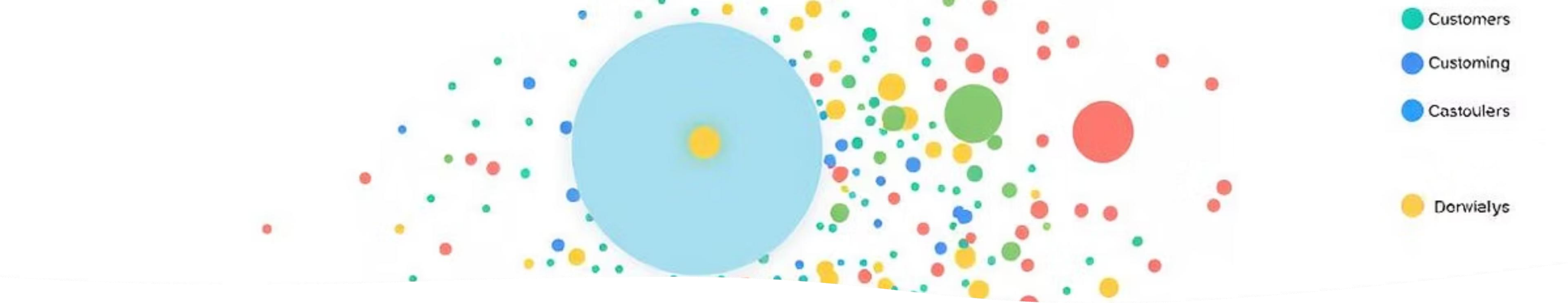
Baseline Clustering Performance

Model	Silhouette	Calinski-Harabasz	Davies-Bouldin
K-means (K=2)	0.215	489.9	1.994
Agglomerative (K=10)	0.148	165.2	1.746
DBSCAN	0.212	56.8	1.112

Key Finding: K-means with K=2 shows best silhouette score, indicating clearer cluster separation

Baseline Model Clustering



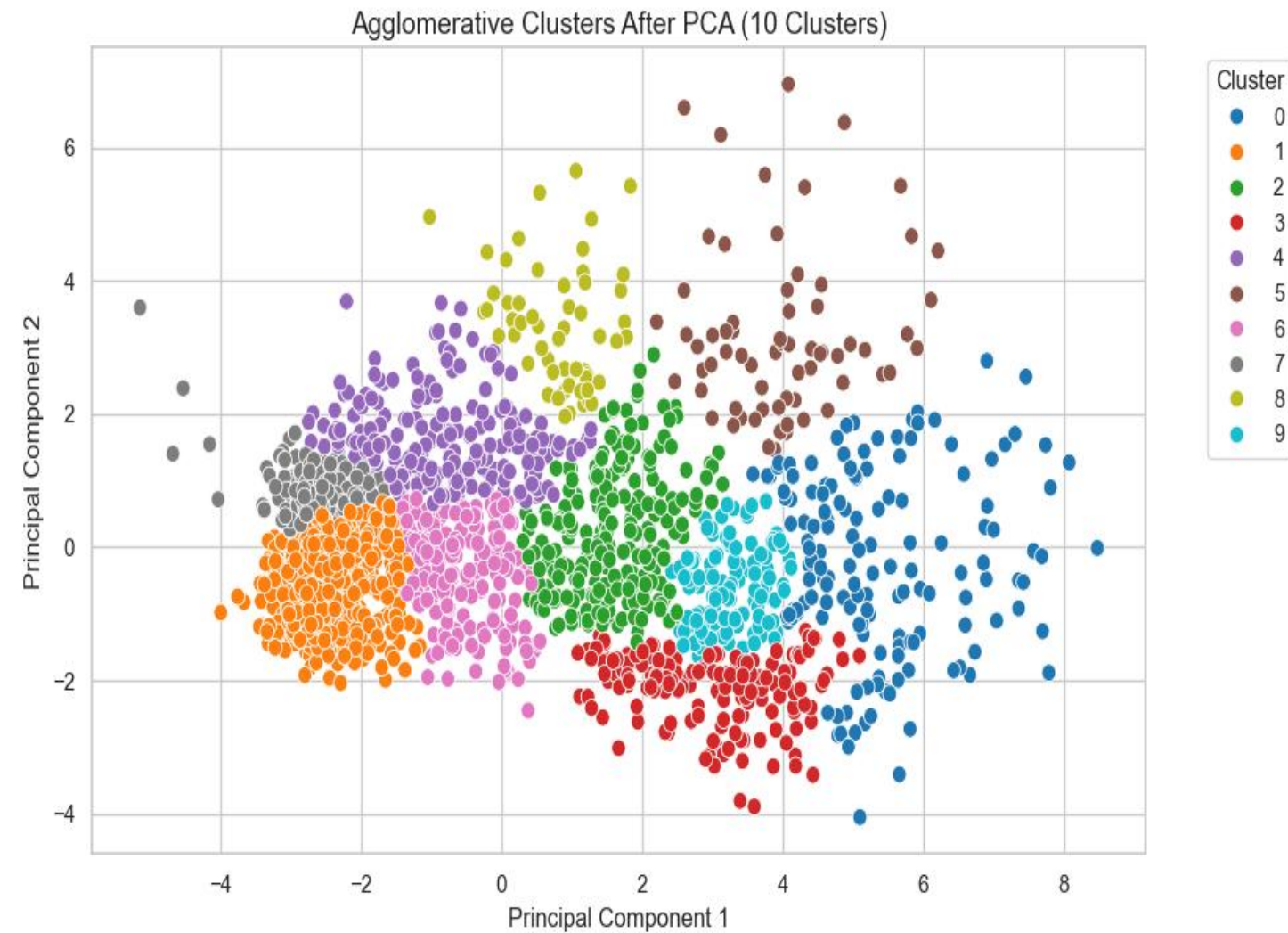
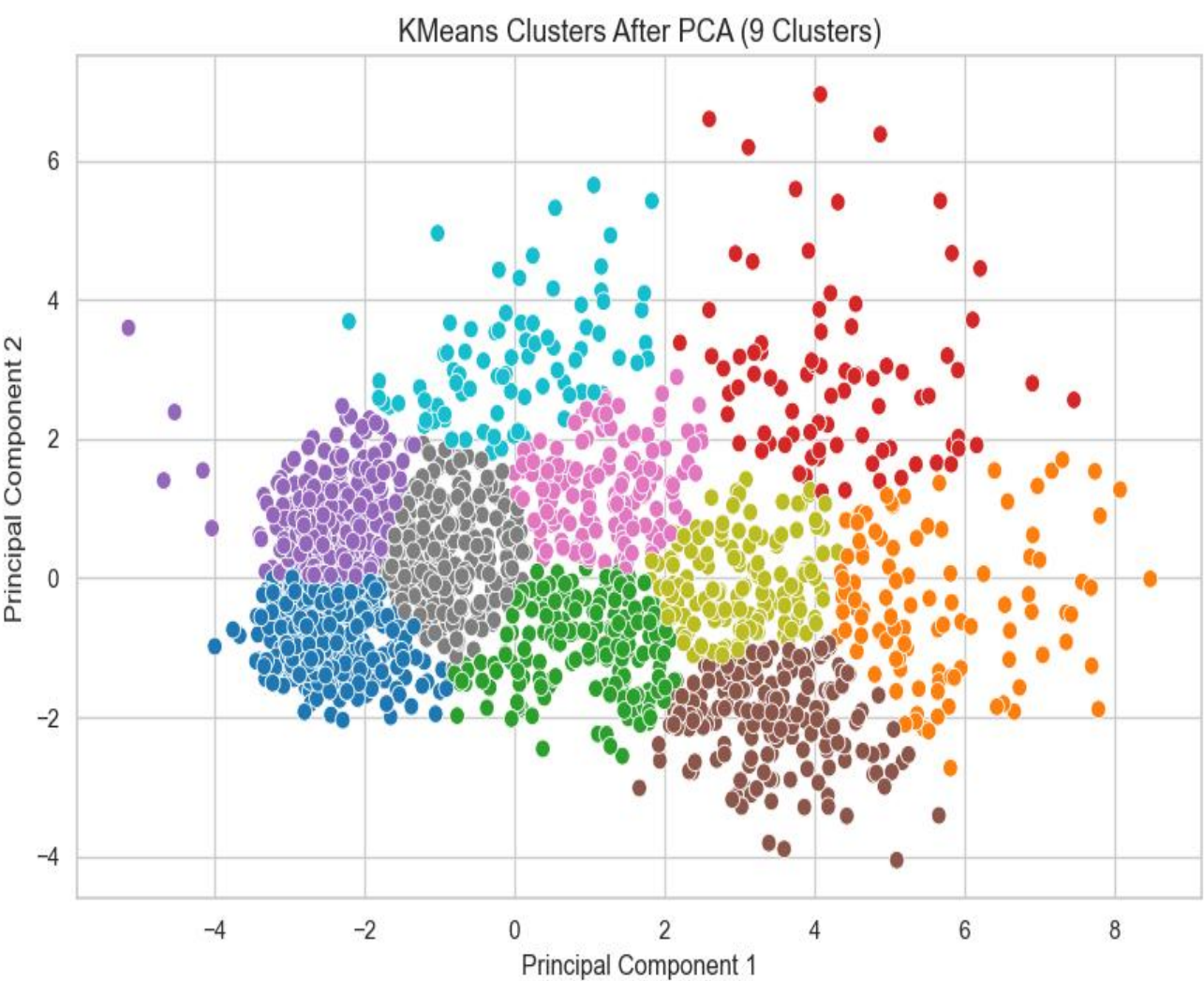


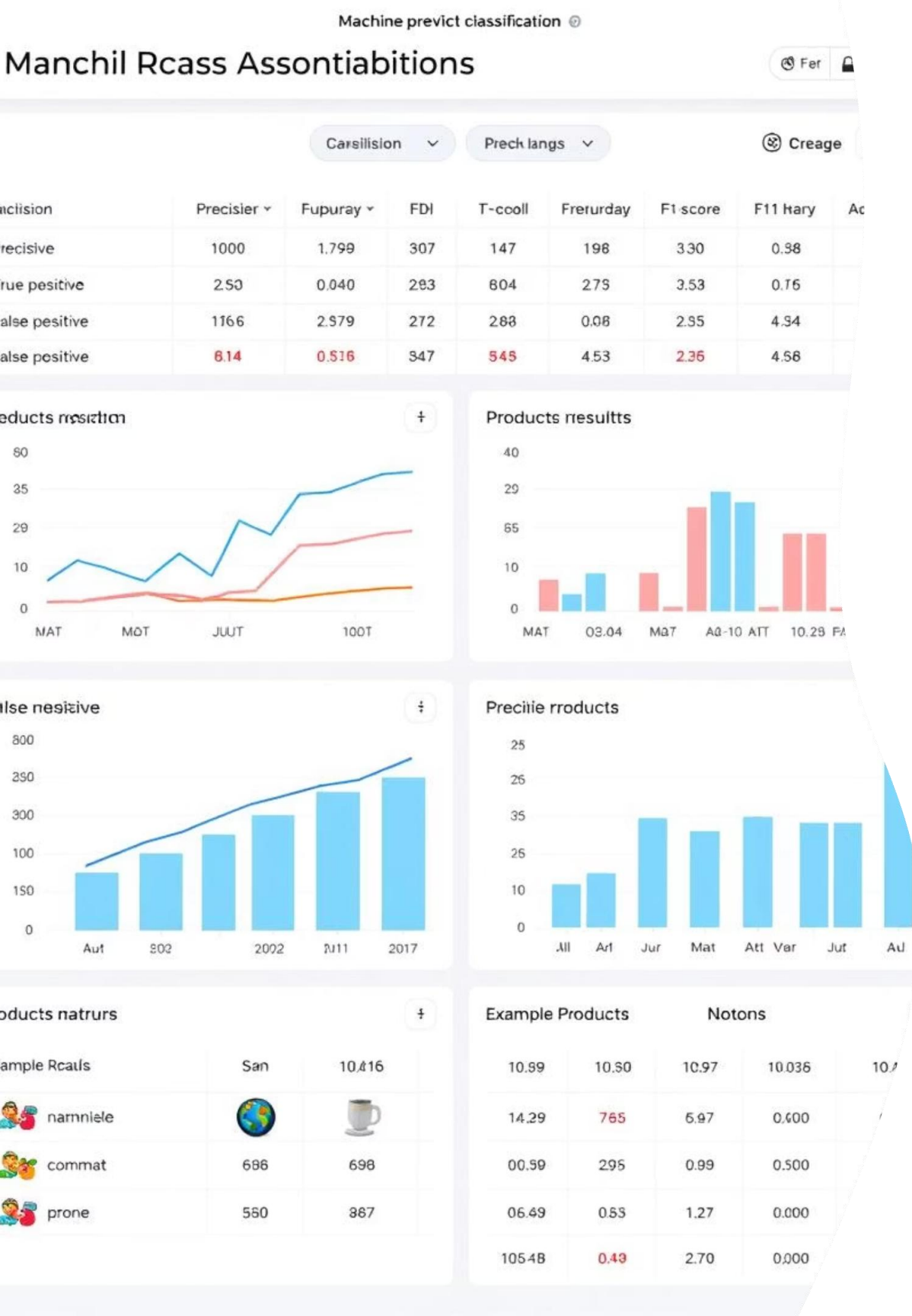
Clustering Performance after PCA

Model	Silhouette	Calinski-Harabasz	Davies-Bouldin
K-means (K=2)	0.37	2420.436	0.84
Agglomerative (K=10)	0.30	2046.964118	0.91
DBSCAN	NaN	NaN	NaN

Key Finding: K-means with K=2 shows best silhouette score, indicating clearer cluster separation

Clustering after PCA





Supervised Learning: Baseline Classification Results

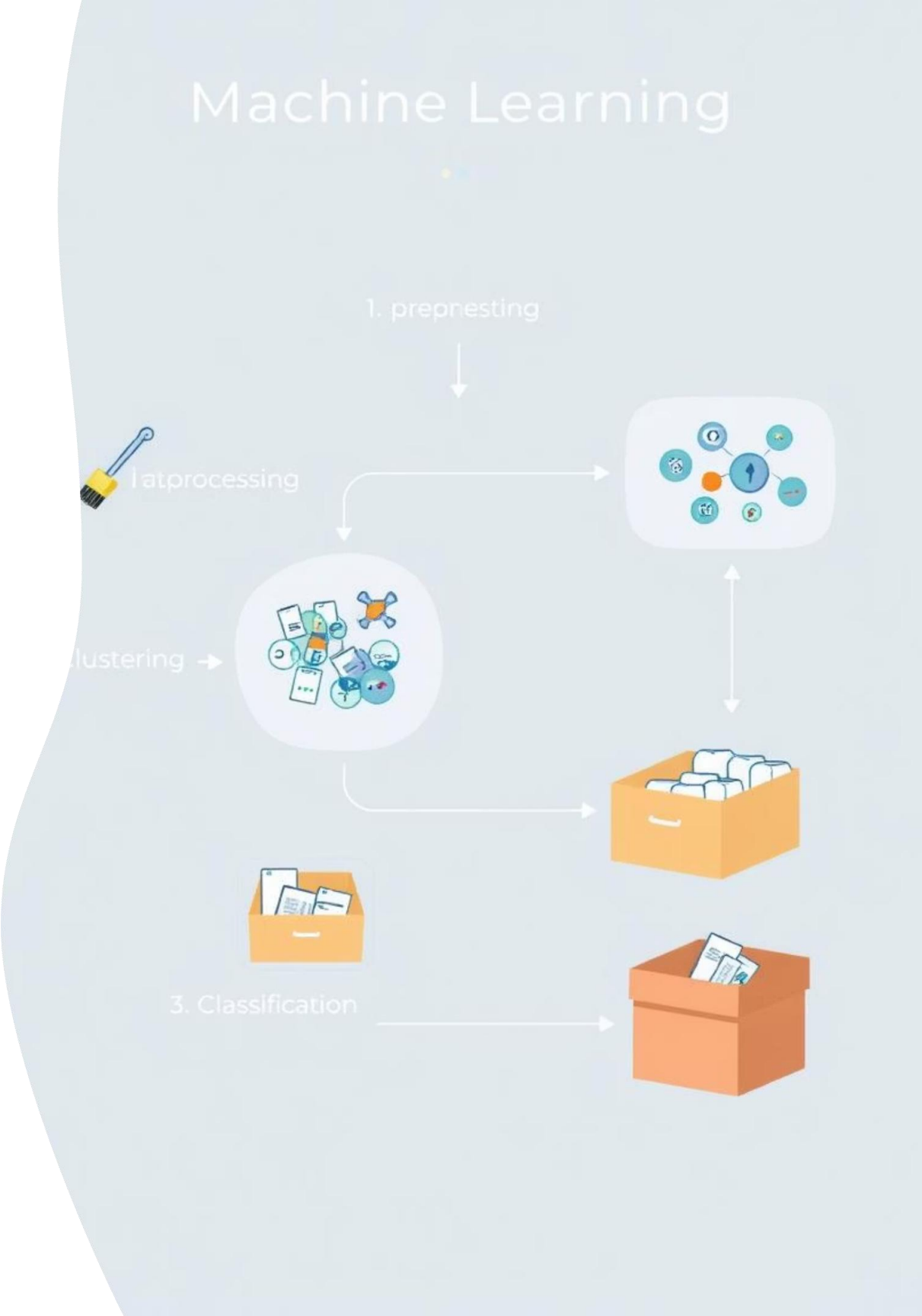
Model Name	Accuracy	Weighted Precision	Weighted Recall	Weighted F1-Score
XGBoost	0.74	0.72	0.74	0.72
Random Forest	0.73	0.72	0.73	0.72
k-NN	0.72	0.68	0.72	0.69
Logistic Regression	0.72	0.63	0.72	0.64
Decision Tree	0.70	0.69	0.70	0.69

Best Performer: XGBoost achieved highest accuracy at 74% with balanced precision-recall

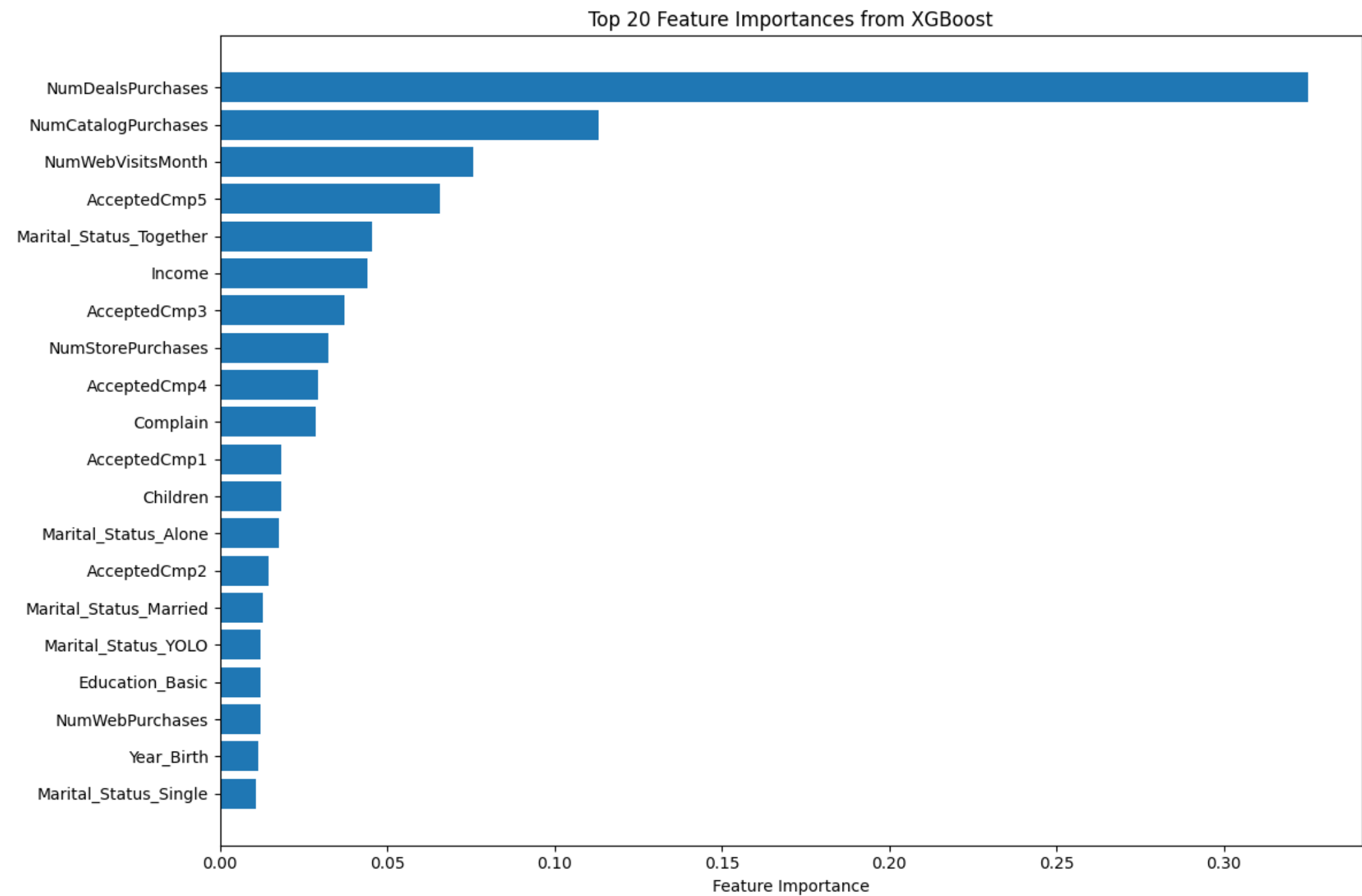
Model Performance Metrics After Applying SMOTE

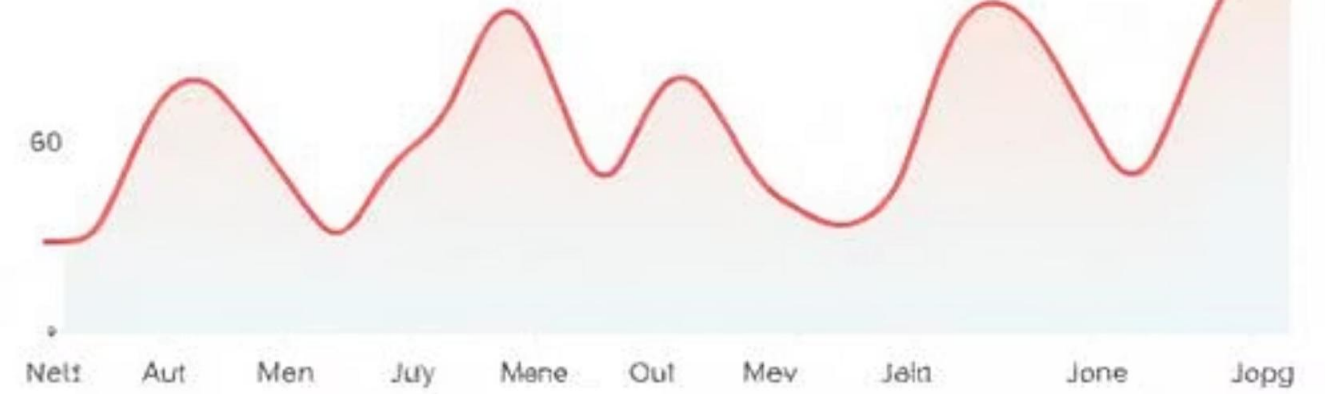
Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1-Score
XGBoost	0.77	0.75	0.77	0.75
Random Forest	0.75	0.72	0.75	0.73
Logistic Regression	0.73	0.67	0.73	0.67
k-NN	0.71	0.67	0.71	0.68
Decision Tree	0.70	0.70	0.70	0.70

Best Performer: XGBoost achieved highest accuracy at 77% with balanced precision-recall



Feature Importance





Dt_customer	Predicted Product	Customer Segment
961	MntWines	2
229	MntMeatProducts	2
1093	MntWines	0
427	MntWines	3
1650	MntFishProducts	2
543	MntWines	3
1801	MntWines	1
1708	MntWines	2
994	MntWines	2
976	MntWines	1

Dashboard Prediction

Dashboard Integration: Real-time customer segmentation combined with product preference predictions enables targeted marketing strategies



Study Limitations

Static Dataset

No time-related variables
preventing modeling of
evolving customer behavior

Missing Psychographics

No brand preferences, lifestyle
choices, or price sensitivity
data

Class Imbalance

Even after SMOTE, some
product classes remained
difficult to predict

Feature Constraints

Available features may not
capture full complexity of
consumer decisions

Business Applications

From Research to Action

Actionable Business Strategies



Persona-Based Segmentation

Create actionable customer personas for **personalized marketing** strategies



Targeted Campaigns

Deploy email/loyalty campaigns based on segment preferences and behaviors



Upselling Opportunities

Target budget-conscious segments with bundle deals and premium offers



Premium Engagement

Cultivate high-income customers through **exclusivity marketing** and VIP programs



Proactive Prediction

Anticipate customer needs before churn, enabling early intervention



Journey Innovation

Build smarter, real-time customer engagement pipelines in modern retail

Future Research Directions

Time-Aware Features

Integrate recency, frequency, and trend analysis for dynamic modeling

Psychographic Data

Include brand loyalty, price sensitivity, and lifestyle preferences

Advanced Models

Explore neural networks, LSTM, and context-aware models for richer predictions

Real-Time Systems

Build operational ML pipelines for live retail environments



Peer Review Feedback

- **Abstract Too Long:** Make it shorter, focus on key contributions only.
- **Literature Review Structure:** Group similar studies, clearly state your innovation.
- **Model Gaps:** Highlight possible reasons for low classification gains (e.g., class imbalance, limited features).
- **DBSCAN Result:** Since it didn't perform well post-PCA, summarize this part briefly.
- **Dashboard to Action:** Connect insights more clearly to business strategies (e.g., campaign design).

Conclusion

0.77

XGBoost Accuracy

Highest prediction accuracy achieved through advanced ensemble methods

10

Customer Segments

Budget buyers, campaign responders, and loyal high spenders identified, Low engagement etc.

The hybrid ML approach enhanced business insight but **behavioral + demographic data alone** may not capture deeper product preferences. Framework provides foundation for future retail intelligence systems.

📌 **Research Impact:** While not yet deployable in real-time retail systems, this unified framework establishes a foundation for next-generation customer intelligence platforms.

References

- Ramkumar, G., Bhuvaneswari, J., Venugopal, S., Kumar, S., Ramasamy, C. K., & Karthick, R. (2025). Enhancing customer segmentation: RFM analysis and K-Means clustering implementation. In *Innovations in Management and Technology* (pp. 109–122). <https://www.taylorfrancis.com/chapters/oa-edit/10.1201/9781003559139-9>
- Bhatia, K., & Gupta, B. P. (2025). Predictive analytics for customer behavior and sales forecasting in retail. *International Journal of World of Sciences*. <https://www.ijwos.com/index.php/home/article/view/10>
- Nasirzonouzi, A. (2025). Leveraging network science for customer segmentation and product recommendation. *Northeast Journal of Complex Systems*, 7(1), Article 3. <https://orb.binghamton.edu/nejcs/vol7/iss1/3/>
- Kabir, M. F., Chowdhury, M. D., Shahiduzzaman, M., & Rana, M. I. C. (2025). Comprehensive customer segmentation and behavior prediction using advanced machine learning and neural network models. *International Journal of Creative Research Thoughts*, 13(1), 478–490.
- Lathifah, S. N., & Azzahra, Z. F. (2025). AI-driven customers segmentation using K-Means clustering. *G-Tech: Jurnal Teknologi Terapan*, 6(1). <https://ejournal.uniramalang.ac.id/index.php/g-tech/article/view/6202>

Thank you for your time!

Knowledge is Power!

Life has given us so many things, what we choose to do with it is what truly matters.

– Dr. Itauma Itauma

Any Questions?

Utilizing data for customer segmentation empowers businesses to predict needs, personalize experiences, and drive meaningful engagement.